

Problem

A series RCL circuit has $R = 30\ \Omega$, $X_C = 50\ \Omega$, and $X_L = 90\ \Omega$. The impedance of the circuit is:

- (a) $30 + j140\ \Omega$
- (b) $30 + j40\ \Omega$
- (c) $30 - j40\ \Omega$
- (d) $-30 - j40\ \Omega$
- (e) $-30 + j40\ \Omega$

Step-by-step solution

Step 1 of 2

Calculate the impedance of the resistor.

$$\begin{aligned} \mathbf{Z}_R &= R \\ &= 30\ \Omega \end{aligned}$$

Calculate the impedance of the inductor.

$$\begin{aligned} \mathbf{Z}_L &= jX_L \\ &= j90\ \Omega \end{aligned}$$

Calculate the impedance of the capacitor.

$$\begin{aligned} \mathbf{Z}_C &= -jX_C \\ &= -j50\ \Omega \end{aligned}$$

Step 2 of 2

Calculate the impedance of the series RLC circuit.

$$\mathbf{Z} = \mathbf{Z}_R + \mathbf{Z}_L + \mathbf{Z}_C$$

Substitute $30\ \Omega$ for $\mathbf{Z}_R$, $j90\ \Omega$ for $\mathbf{Z}_L$ and $-j50\ \Omega$ for $\mathbf{Z}_C$.

$$\begin{aligned} \mathbf{Z} &= 30 + j90 - j50 \\ &= 30 + j40\ \Omega \end{aligned}$$

Thus, the correct answer is (b).